

# 365 Data Science SQL Basics – Topic 2.1: SELECT & WHERE Filtering

Exhaustive Notes on Data Querying with SELECT, FROM, WHERE & Comparison Operators

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## 1. The SELECT Statement & Column Retrieval

The `SELECT` statement retrieves data records from relational tables. Using `SELECT \*` retrieves all columns, whereas specifying explicit column names (`SELECT col1, col2`) optimizes query memory usage.

### SQL SELECT Query Logical Order of Execution



*Diagram: SQL SELECT Query Logical Order of Execution.*

## 2. The WHERE Clause & Comparison Operators

The `WHERE` clause filters raw table rows based on specified logical conditions (`=`, `<>`, `>`, `<`, `>=`, `<<=`, `<`).

```
SELECT * FROM employees
WHERE gender = 'F' AND hire_date >= '2020-01-01';
```

## 3. Topic 2.1 Viva Q&A

1. Q: What is the purpose of the WHERE clause? A: Filters rows before any grouping or aggregation takes place.